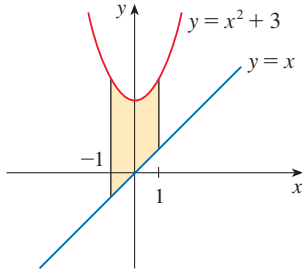


7.1 AREAS BETWEEN CURVES

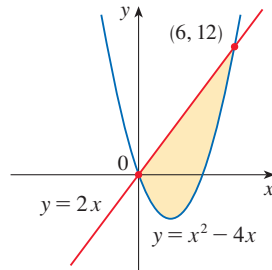
A Click here for answers.

1–4 ■ Find the area of the shaded region.

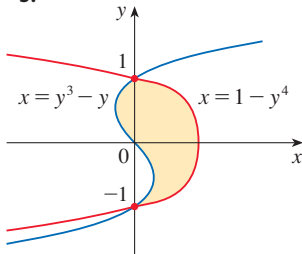
1.



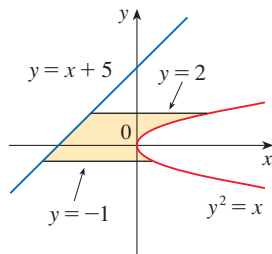
2.



3.



4.



5–10 ■ Sketch the region enclosed by the given curves. Decide whether to integrate with respect to x or y . Draw a typical approximating rectangle and label its height and width. Then find the area of the region.

5. $y = 4x^2$, $y = x^2 + 3$

6. $y = x + 1$, $y = (x - 1)^2$, $x = -1$, $x = 2$

7. $y = x^2 + 1$, $y = 3 - x^2$, $x = -2$, $x = 2$

8. $y^2 = x$, $x - 2y = 3$

9. $y = 1/x$, $x = 0$, $y = 1$, $y = 2$

10. $y = \cos x$, $y = \sec^2 x$, $x = -\pi/4$, $x = \pi/4$

11–36 ■ Sketch the region bounded by the given curves and find the area of the region.

11. $y = x$, $y = x^3$

12. $y = \sqrt{x}$, $y = x/2$

13. $y = \sqrt{x - 1}$, $x - 3y + 1 = 0$

14. $y = x^4 - x^2$, $y = 1 - x^2$

15. $y = x^2 + 2$, $y = 2x + 5$, $x = 0$, $x = 6$

16. $x + y^2 = 2$, $x + y = 0$

17. $y = x^2 + 3$, $y = x$, $x = -1$, $x = 1$

S Click here for solutions.

18. $y = x^4$, $y = -x - 1$, $x = -2$, $x = 0$

19. $y^2 = x$, $y = x + 5$, $y = -1$, $y = 2$

20. $x + y^2 = 0$, $x = y^2 + 1$, $y = 0$, $y = 3$

21. $y = x^2 - 4x$, $y = 2x$

22. $x^2 + 2x + y = 0$, $x + y + 2 = 0$

23. $y = 4 - x^2$, $y = x + 2$, $x = -3$, $x = 0$

24. $y = x^2 + 2x + 2$, $y = x + 4$, $x = -3$, $x = 2$

25. $y = x^3 - 4x^2 + 3x$, $y = x^2 - x$

26. $y = x$, $y = \sin x$, $x = -\pi/4$, $x = \pi/2$

27. $y = \sin x$, $y = \cos 2x$, $x = 0$, $x = \pi/4$

28. $y = |x|$, $y = (x + 1)^2 - 7$, $x = -4$

29. $y = |x - 1|$, $y = x^2 - 3$, $x = 0$

30. $x = 3y$, $x + y = 0$, $7x + 3y = 24$

31. $y = x\sqrt{1 - x^2}$, $y = x - x^3$

32. $y = 1/x$, $y = 1/x^2$, $x = 1$, $x = 2$

33. $y = x^2$, $y = 2/(x^2 + 1)$

34. $y = 2^x$, $y = 5^x$, $x = -1$, $x = 1$

35. $y = e^x$, $y = e^{3x}$, $x = 1$

36. $y = e^x$, $y = e^{-x}$, $x = -2$, $x = 1$

37. Evaluate

$$\int_0^{\pi} \left| \sin x - \frac{2}{\pi} x \right| dx$$

and interpret it as the area of a region. Sketch the region.

38–39 ■ Use the Midpoint Rule with $n = 4$ to approximate the area of the region bounded by the given curves.

38. $y = \sqrt{1 + x^3}$, $y = 1 - x$, $x = 2$

39. $y = x \tan x$, $y = x$



40–41 ■ Use a graph to find approximate x -coordinates of the points of intersection of the given curves. Then use the Midpoint Rule with $n = 4$ to approximate the area of the region bounded by the curves.

40. $y = 1 + 3x - 2x^2$, $y = \sqrt{1 + x^4}$

41. $y = x^2 - x$, $y = \sin(x^2)$

42–43 ■ Find the area of the region bounded by the given curves by two methods: (a) integrating with respect to x , and (b) integrating with respect to y .

42. $4x + y^2 = 0$, $y = 2x + 4$

43. $x + 1 = 2(y - 2)^2$, $x + 6y = 7$

44–45 ■ Use calculus to find the area of the triangle with the given vertices.

44. $(0, 0), (1, 8), (4, 3)$

45. $(-2, 5), (0, -3), (5, 2)$

A line graph on a grid with 4 vertical and 3 horizontal lines. The line starts at the bottom-left intersection, rises to the top-left intersection, dips to the top-right intersection, and then rises to the top-right intersection.

 **46–48** ■ Use a graph to find approximate x -coordinates of the points of intersection of the given curves. Then find (approximately) the area of the region bounded by the curves.

46. $y = \sqrt{x+1}$, $y = x^2$

47. $y = x^4 - 1$, $y = x \sin(x^2)$

48. $y = x^2$, $y = e^{-x^2}$

7.1 ANSWERS**E** [Click here for exercises.](#)

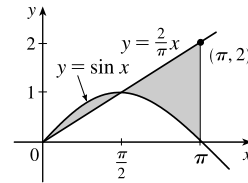
1. $\frac{20}{3}$
2. 36
3. $\frac{8}{5}$
4. $\frac{33}{2}$
5. 4
6. $\frac{31}{6}$
7. 8
8. $\frac{32}{3}$
9. $\ln 2$
10. $2 - \sqrt{2}$
11. $\frac{1}{2}$
12. $\frac{4}{3}$
13. $\frac{1}{6}$
14. $\frac{8}{5}$
15. 36
16. $\frac{9}{2}$
17. $\frac{20}{3}$
18. $\frac{32}{5}$
19. $\frac{33}{2}$
20. 21
21. 36
22. $\frac{9}{2}$
23. $\frac{31}{6}$
24. $\frac{49}{6}$
25. $\frac{71}{6}$
26. $\frac{5}{32}\pi^2 + \frac{1}{\sqrt{2}} - 2$
27. $\frac{1}{2}(3\sqrt{3} - \sqrt{2} - 3)$
28. 34
29. $\frac{13}{3}$
30. 12
31. $\frac{1}{6}$
32. $\ln 2 - \frac{1}{2}$
33. $\pi - \frac{2}{3}$
34. $\frac{16}{5 \ln 5} - \frac{1}{2 \ln 2}$

S [Click here for solutions.](#)

35. $\frac{1}{3}e^3 - e + \frac{2}{3}$

36. $e^2 + e + e^{-1} + e^{-2} - 4$

37. $\frac{\pi}{2}$



38. 3.22

39. 0.13

40. 0.83

41. 0.81

42. 9

43. $\frac{1}{3}$

44. $\frac{29}{2}$

45. 25

46. 1.38

47. 1.78

48. 0.98

7.1 SOLUTIONS

Click here for exercises.

$$1. A = \int_{-1}^1 [(x^2 + 3) - x] dx = 2 \int_0^1 (x^2 + 3) dx$$

[by Theorem 4.5.6(a)]

$$= 2 \left[\frac{1}{3}x^3 + 3x \right]_0^1 = 2 \left(\frac{1}{3} + 3 \right) = \frac{20}{3}$$

$$2. A = \int_0^6 [2x - (x^2 - 4x)] dx = \int_0^6 (6x - x^2) dx$$

$$= \left[3x^2 - \frac{1}{3}x^3 \right]_0^6 = 108 - 72 = 36$$

$$3. A = \int_{-1}^1 [(1 - y^4) - (y^3 - y)] dy = 2 \int_0^1 (1 - y^4) dy$$

[by Theorem 4.5.6(a)]

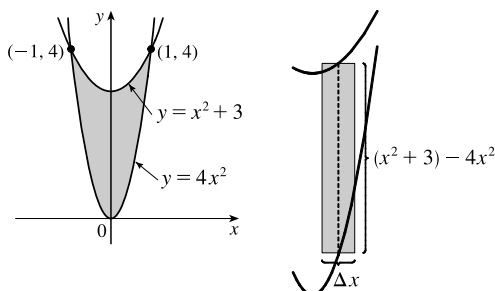
$$= 2 \left[-\frac{1}{5}y^5 + y \right]_0^1 = 2 \left(-\frac{1}{5} + 1 \right) = \frac{8}{5}$$

$$4. A = \int_{-1}^2 [y^2 - (y - 5)] dy = \left[\frac{1}{3}y^3 - \frac{1}{2}y^2 + 5y \right]_{-1}^2$$

$$= \left(\frac{8}{3} - 2 + 10 \right) - \left(-\frac{1}{3} - \frac{1}{2} - 5 \right) = \frac{33}{2}$$

$$5. A = \int_{-1}^1 [(x^2 + 3) - 4x^2] dx = 2 \int_0^1 (3 - 3x^2) dx$$

$$= 2 \left[3x - x^3 \right]_0^1 = 2(3 - 1) = 4$$



$$6. x + 1 = (x - 1)^2 \Rightarrow x + 1 = x^2 - 2x + 1 \Rightarrow 0 = x^2 - 3x \Rightarrow 0 = x(x - 3) \Rightarrow x = 0 \text{ or } 3.$$

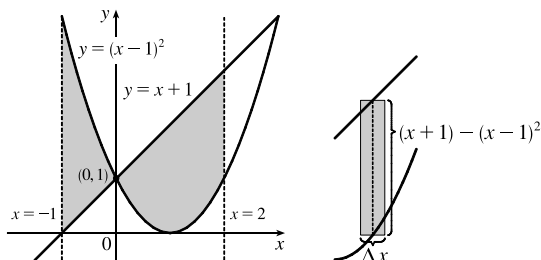
$$A = \int_{-1}^2 |(x + 1) - (x - 1)^2| dx$$

$$= \int_{-1}^0 [(x - 1)^2 - (x + 1)] dx + \int_0^2 [(x + 1) - (x - 1)^2] dx$$

$$= \int_{-1}^0 (x^2 - 2x + 1 - x - 1) dx + \int_0^2 (x^2 + 2x + 1 - x^2 + 2x - 1) dx$$

$$= \left[\frac{1}{3}x^3 - \frac{3}{2}x^2 + x \right]_{-1}^0 + \left[\frac{2}{3}x^3 + 2x^2 \right]_0^2$$

$$= 0 - \left(-\frac{1}{3} - \frac{3}{2} + 1 \right) + \left(\frac{16}{3} + 8 \right) - 0 = \frac{31}{6}$$



$$7. A = \int_{-2}^{-1} [(x^2 + 1) - (3 - x^2)] dx$$

$$+ \int_{-1}^1 [(3 - x^2) - (x^2 + 1)] dx$$

$$+ \int_1^2 [(x^2 + 1) - (3 - x^2)] dx$$

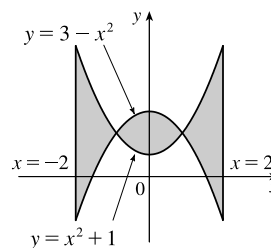
$$= \int_{-2}^{-1} (2x^2 - 2) dx + \int_{-1}^1 (2 - 2x^2) dx$$

$$+ \int_1^2 (2x^2 - 2) dx$$

$$= 2 \int_0^1 (2 - 2x^2) dx + 2 \int_1^2 (2x^2 - 2) dx \text{ [symmetry]}$$

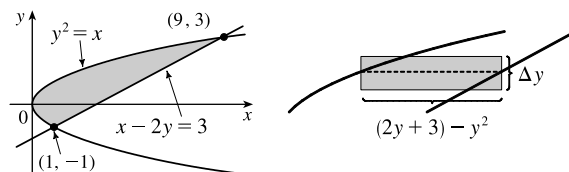
$$= 2 \left[2x - \frac{2}{3}x^3 \right]_0^1 + 2 \left[\frac{2}{3}x^3 - 2x \right]_1^2$$

$$= 2 \left(2 - \frac{2}{3} \right) + 2 \left(\frac{16}{3} - 4 \right) - 2 \left(\frac{2}{3} - 2 \right) = 8$$

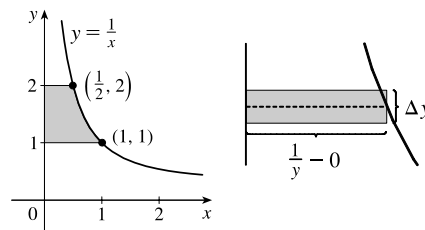


$$8. A = \int_{-1}^3 [(2y + 3) - y^2] dy = \left[y^2 + 3y - \frac{1}{3}y^3 \right]_{-1}^3$$

$$= (9 + 9 - 9) - \left(1 - 3 + \frac{1}{3} \right) = \frac{32}{3}$$



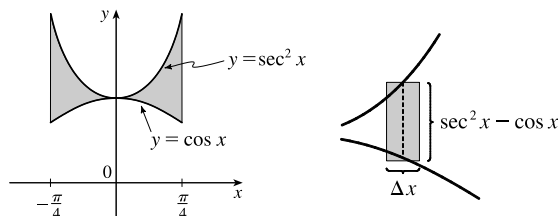
$$9. A = \int_1^2 (1/y) dy = [\ln y]_1^2 = \ln 2 - \ln 1 = \ln 2 \approx 0.69$$



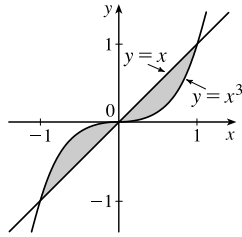
$$10. A = \int_{-\pi/4}^{\pi/4} (\sec^2 x - \cos x) dx$$

$$= 2 \int_0^{\pi/4} (\sec^2 x - \cos x) dx = 2 [\tan x - \sin x]_0^{\pi/4}$$

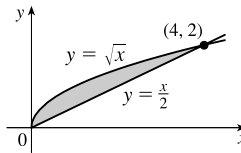
$$= 2 \left(1 - \frac{1}{\sqrt{2}} \right) = 2 - \sqrt{2} \approx 0.59$$



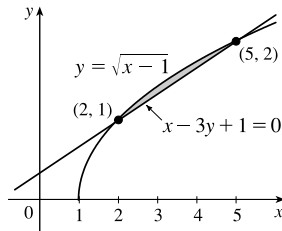
$$\begin{aligned}
 11. A &= \int_{-1}^0 (x^3 - x) dx + \int_0^1 (x - x^3) dx \\
 &= 2 \int_0^1 (x - x^3) dx = 2 \left[\frac{1}{2}x^2 - \frac{1}{4}x^4 \right]_0^1 \\
 &= 2 \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{1}{2}
 \end{aligned}$$



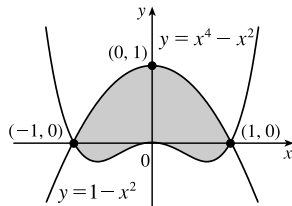
$$\begin{aligned}
 12. A &= \int_0^4 \left(\sqrt{x} - \frac{1}{2}x \right) dx = \left[\frac{2}{3}x^{3/2} - \frac{1}{4}x^2 \right]_0^4 \\
 &= \left(\frac{16}{3} - 4 \right) - 0 = \frac{4}{3}
 \end{aligned}$$



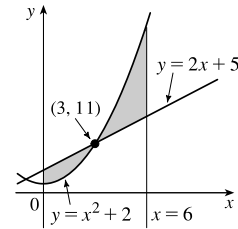
$$\begin{aligned}
 13. A &= \int_2^5 \left(\sqrt{x-1} - \frac{1}{3}x - \frac{1}{3} \right) dx \\
 &= \left[\frac{2}{3}(x-1)^{3/2} - \frac{1}{6}x^2 - \frac{1}{3}x \right]_2^5 \\
 &= \left(\frac{16}{3} - \frac{25}{6} - \frac{5}{3} \right) - \left(\frac{2}{3} - \frac{4}{6} - \frac{2}{3} \right) = \frac{1}{6}
 \end{aligned}$$



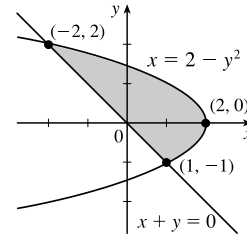
$$\begin{aligned}
 14. A &= \int_{-1}^1 [(1 - x^2) - (x^4 - x^2)] dx \\
 &= 2 \int_0^1 (1 - x^4) dx = 2 \left[x - \frac{1}{5}x^5 \right]_0^1 \\
 &= 2 \left(1 - \frac{1}{5} \right) = \frac{8}{5}
 \end{aligned}$$



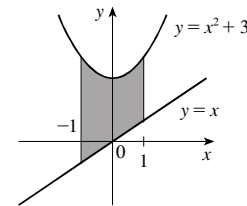
$$\begin{aligned}
 15. A &= \int_0^3 [(2x + 5) - (x^2 + 2)] dx \\
 &\quad + \int_3^6 [(x^2 + 2) - (2x + 5)] dx \\
 &= \int_0^3 (-x^2 + 2x + 3) dx + \int_3^6 (x^2 - 2x - 3) dx \\
 &= \left[-\frac{1}{3}x^3 + x^2 + 3x \right]_0^3 + \left[\frac{1}{3}x^3 - x^2 - 3x \right]_3^6 \\
 &= (-9 + 9 + 9) - 0 + (72 - 36 - 18) - (9 - 9 - 9) \\
 &= 36
 \end{aligned}$$



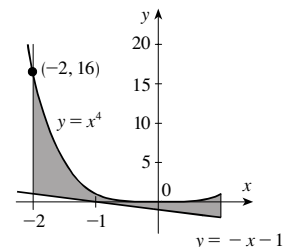
$$\begin{aligned}
 16. A &= \int_{-1}^2 [2 - y^2 - (-y)] dy = \int_{-1}^2 (-y^2 + y + 2) dy \\
 &= \left[-\frac{1}{3}y^3 + \frac{1}{2}y^2 + 2y \right]_{-1}^2 \\
 &= \left(-\frac{8}{3} + 2 + 4 \right) - \left(\frac{1}{3} + \frac{1}{2} - 2 \right) = \frac{9}{2}
 \end{aligned}$$



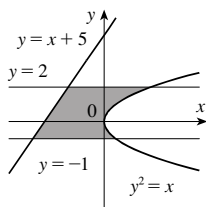
$$\begin{aligned}
 17. A &= \int_{-1}^1 [(x^2 + 3) - x] dx = \int_{-1}^1 (x^2 - x + 3) dx \\
 &= \left[\frac{1}{3}x^3 - \frac{1}{2}x^2 + 3x \right]_{-1}^1 \\
 &= \left(\frac{1}{3} - \frac{1}{2} + 3 \right) - \left(-\frac{1}{3} - \frac{1}{2} - 3 \right) = \frac{20}{3}
 \end{aligned}$$



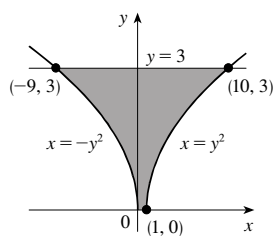
$$\begin{aligned}
 18. A &= \int_{-2}^0 [x^4 - (-x - 1)] dx = \int_{-2}^0 (x^4 + x + 1) dx \\
 &= \left[\frac{1}{5}x^5 + \frac{1}{2}x^2 + x \right]_{-2}^0 = 0 - \left(-\frac{32}{5} + 2 - 2 \right) \\
 &= \frac{32}{5}
 \end{aligned}$$



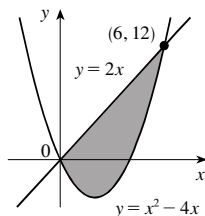
$$\begin{aligned}
 19. A &= \int_{-1}^2 [y^2 - (y - 5)] dy = \left[\frac{1}{3}y^3 - \frac{1}{2}y^2 + 5y \right]_{-1}^2 \\
 &= \left(\frac{8}{3} - 2 + 10 \right) - \left(-\frac{1}{3} - \frac{1}{2} - 5 \right) = \frac{33}{2}
 \end{aligned}$$



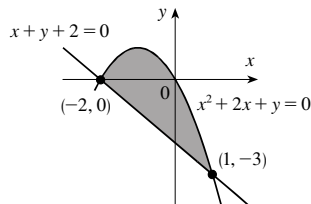
$$\begin{aligned}
 20. A &= \int_0^3 [(y^2 + 1) - (-y^2)] dy = \int_0^3 (2y^2 + 1) dy \\
 &= \left[\frac{2}{3}y^3 + y \right]_0^3 = 18 + 3 = 21
 \end{aligned}$$



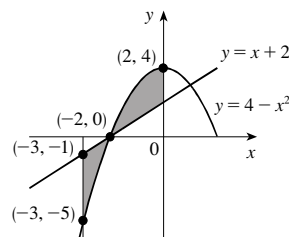
$$\begin{aligned}
 21. A &= \int_0^6 [2x - (x^2 - 4x)] dx = \int_0^6 (6x - x^2) dx \\
 &= \left[3x^2 - \frac{1}{3}x^3 \right]_0^6 = 108 - 72 = 36
 \end{aligned}$$



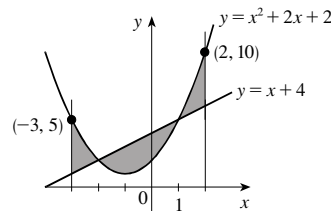
$$\begin{aligned}
 22. A &= \int_{-2}^1 [-x^2 - 2x - (-x - 2)] dx \\
 &= \int_{-2}^1 (-x^2 - x + 2) dx = \left[-\frac{1}{3}x^3 - \frac{1}{2}x^2 + 2x \right]_{-2}^1 \\
 &= \left(-\frac{1}{3} - \frac{1}{2} + 2 \right) - \left(\frac{8}{3} - 2 - 4 \right) = \frac{9}{2}
 \end{aligned}$$



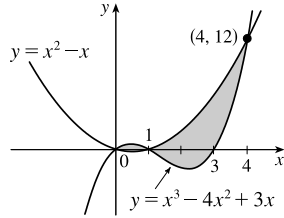
$$\begin{aligned}
 23. A &= \int_{-3}^0 |(4 - x^2) - (x + 2)| dx \\
 &= \int_{-3}^{-2} [(x + 2) - (4 - x^2)] dx \\
 &\quad + \int_{-2}^0 [(4 - x^2) - (x + 2)] dx \\
 &= \int_{-3}^{-2} (x^2 + x - 2) dx + \int_{-2}^0 (-x^2 - x + 2) dx \\
 &= \left[\frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x \right]_{-3}^{-2} \\
 &\quad + \left[-\frac{1}{3}x^3 - \frac{1}{2}x^2 + 2x \right]_{-2}^0 \\
 &= \left(-\frac{8}{3} + 2 + 4 \right) - \left(-9 + \frac{9}{2} + 6 \right) \\
 &\quad + 0 - \left(\frac{8}{3} - 2 - 4 \right) = \frac{31}{6}
 \end{aligned}$$



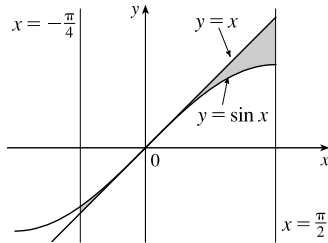
$$\begin{aligned}
 24. A &= \int_{-3}^{-2} [(x^2 + 2x + 2) - (x + 4)] dx \\
 &\quad + \int_{-2}^1 [(x + 4) - (x^2 + 2x + 2)] dx \\
 &\quad + \int_1^2 [(x^2 + 2x + 2) - (x + 4)] dx \\
 &= \int_{-3}^{-2} (x^2 + x - 2) dx + \int_{-2}^1 (-x^2 - x + 2) dx \\
 &\quad + \int_1^2 (x^2 + x - 2) dx \\
 &= \left[\frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x \right]_{-3}^{-2} + \left[-\frac{1}{3}x^3 - \frac{1}{2}x^2 + 2x \right]_{-2}^1 \\
 &\quad + \left[\frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x \right]_1^2 \\
 &= \left[\left(-\frac{8}{3} + 2 + 4 \right) - \left(-9 + \frac{9}{2} + 6 \right) \right] \\
 &\quad + \left[\left(-\frac{1}{3} - \frac{1}{2} + 2 \right) - \left(\frac{8}{3} - 2 - 4 \right) \right] \\
 &\quad + \left[\left(\frac{8}{3} + 2 - 4 \right) - \left(\frac{1}{3} + \frac{1}{2} - 2 \right) \right] = \frac{49}{6}
 \end{aligned}$$



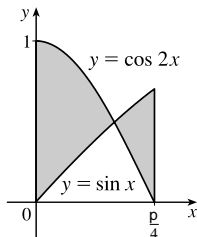
$$\begin{aligned}
 25. A &= \int_0^1 [(x^3 - 4x^2 + 3x) - (x^2 - x)] dx \\
 &\quad + \int_1^4 [x^2 - x - (x^3 - 4x^2 + 3x)] dx \\
 &= \int_0^1 (x^3 - 5x^2 + 4x) dx + \int_1^4 (-x^3 + 5x^2 - 4x) dx \\
 &= \left[\frac{1}{4}x^4 - \frac{5}{3}x^3 + 2x^2 \right]_0^1 + \left[-\frac{1}{4}x^4 + \frac{5}{3}x^3 - 2x^2 \right]_1^4 \\
 &= \left(\frac{1}{4} - \frac{5}{3} + 2 \right) - 0 + \left(-64 + \frac{320}{3} - 32 \right) \\
 &\quad - \left(-\frac{1}{4} + \frac{5}{3} - 2 \right) = \frac{71}{6}
 \end{aligned}$$



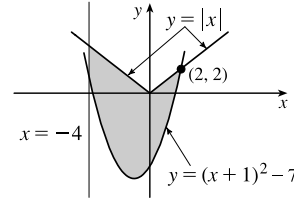
$$\begin{aligned}
 26. A &= \int_{-\pi/4}^0 (\sin x - x) dx + \int_0^{\pi/2} (x - \sin x) dx \\
 &= \left[-\cos x - \frac{1}{2}x^2 \right]_{-\pi/4}^0 + \left[\frac{1}{2}x^2 + \cos x \right]_0^{\pi/2} \\
 &= 1 - \left(-\frac{1}{\sqrt{2}} - \frac{\pi^2}{32} \right) + \frac{\pi^2}{8} - 1 = \frac{5}{32}\pi^2 + \frac{1}{\sqrt{2}} - 2
 \end{aligned}$$



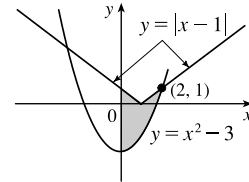
$$\begin{aligned}
 27. \sin x &= \cos 2x = 1 - 2\sin^2 x \Leftrightarrow \\
 2\sin^2 x + \sin x - 1 &= 0 \Leftrightarrow \\
 (2\sin x - 1)(\sin x + 1) &= 0 \Leftrightarrow \sin x = \frac{1}{2} \text{ or } -1 \Leftrightarrow \\
 x &= \frac{\pi}{6}. \\
 A &= \int_0^{\pi/6} (\cos 2x - \sin x) dx + \int_{\pi/6}^{\pi/4} (\sin x - \cos 2x) dx \\
 &= \left[\frac{1}{2}\sin 2x + \cos x \right]_0^{\pi/6} - \left[\frac{1}{2}\sin 2x + \cos x \right]_{\pi/6}^{\pi/4} \\
 &= \left(\frac{1}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right) - 1 - 1 \left(\frac{1}{2} + \frac{\sqrt{2}}{2} \right) + \left(\frac{1}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right) \\
 &= \frac{1}{2}(3\sqrt{3} - \sqrt{2} - 3)
 \end{aligned}$$



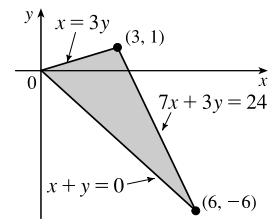
$$\begin{aligned}
 28. A &= \int_{-4}^0 (-x - [(x+1)^2 - 7]) dx \\
 &\quad + \int_0^2 (x - [(x+1)^2 - 7]) dx \\
 &= \int_{-4}^0 (-x^2 - 3x + 6) dx + \int_0^2 (-x^2 - x + 6) dx \\
 &= \left[-\frac{1}{3}x^3 - \frac{3}{2}x^2 + 6x \right]_{-4}^0 + \left[-\frac{1}{3}x^3 - \frac{1}{2}x^2 + 6x \right]_0^2 \\
 &= 0 - \left(\frac{64}{3} - 24 - 24 \right) + \left(-\frac{8}{3} - 2 + 12 \right) - 0 = 34
 \end{aligned}$$



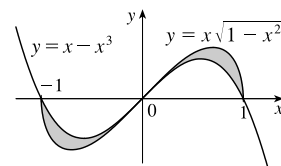
$$\begin{aligned}
 29. A &= \int_0^1 [(1-x) - (x^2 - 3)] dx \\
 &\quad + \int_1^2 [(x-1) - (x^2 - 3)] dx \\
 &= \left[-\frac{1}{3}x^3 - \frac{1}{2}x^2 + 4x \right]_0^1 + \left[-\frac{1}{3}x^3 + \frac{1}{2}x^2 + 2x \right]_1^2 \\
 &= \left(-\frac{1}{3} - \frac{1}{2} + 4 \right) - 0 + \left(-\frac{8}{3} + 2 + 4 \right) \\
 &\quad - \left(-\frac{1}{3} + \frac{1}{2} + 2 \right) = \frac{13}{3}
 \end{aligned}$$



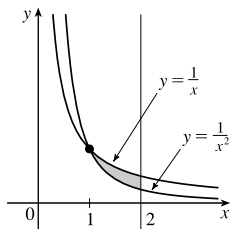
$$\begin{aligned}
 30. A &= \int_0^3 \left[\frac{1}{3}x - (-x) \right] dx + \int_3^6 \left[\left(8 - \frac{7}{3}x \right) - (-x) \right] dx \\
 &= \int_0^3 \frac{4}{3}x dx + \int_3^6 \left(-\frac{4}{3}x + 8 \right) dx \\
 &= \left[\frac{2}{3}x^2 \right]_0^3 + \left[-\frac{2}{3}x^2 + 8x \right]_3^6 \\
 &= (6 - 0) + (24 - 18) = 12
 \end{aligned}$$



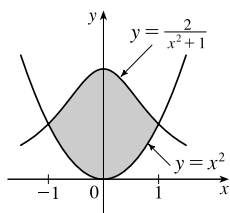
$$\begin{aligned}
 31. A &= \int_{-1}^1 |x\sqrt{1-x^2} - (x-x^3)| dx \\
 &= \int_{-1}^0 (x-x^3 - x\sqrt{1-x^2}) dx \\
 &\quad + \int_0^1 (x\sqrt{1-x^2} - x + x^3) dx \\
 &= 2 \int_0^1 (x\sqrt{1-x^2} - x + x^3) dx \quad (\text{by symmetry}) \\
 &= 2 \left[-\frac{1}{3}(1-x^2)^{3/2} - \frac{1}{2}x^2 + \frac{1}{4}x^4 \right]_0^1 \\
 &= 2 \left[\left(-\frac{1}{2} + \frac{1}{4} \right) + \frac{1}{3} \right] = \frac{1}{6}
 \end{aligned}$$



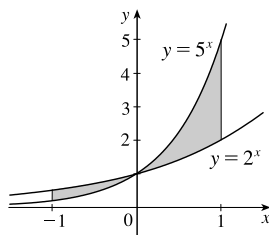
$$\begin{aligned}
 32. A &= \int_1^2 \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = \left[\ln x + \frac{1}{x} \right]_1^2 \\
 &= (\ln 2 + \tfrac{1}{2}) - (\ln 1 + 1) = \ln 2 - \tfrac{1}{2}
 \end{aligned}$$



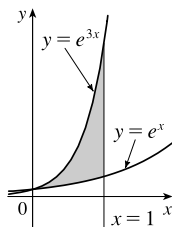
$$\begin{aligned}
 33. A &= 2 \int_0^1 \left(\frac{2}{x^2 + 1} - x^2 \right) dx = \left[4 \tan^{-1} x - \frac{2}{3} x^3 \right]_0^1 \\
 &= 4 \cdot \frac{\pi}{4} - \frac{2}{3} = \pi - \frac{2}{3}
 \end{aligned}$$



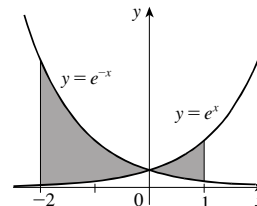
$$\begin{aligned}
 34. A &= \int_{-1}^0 (2^x - 5^x) dx + \int_0^1 (5^x - 2^x) dx \\
 &= \left[\frac{2^x}{\ln 2} - \frac{5^x}{\ln 5} \right]_{-1}^0 + \left[\frac{5^x}{\ln 5} - \frac{2^x}{\ln 2} \right]_0^1 \\
 &= \left(\frac{1}{\ln 2} - \frac{1}{\ln 5} \right) - \left(\frac{1/2}{\ln 2} - \frac{1/5}{\ln 5} \right) \\
 &\quad + \left(\frac{5}{\ln 5} - \frac{2}{\ln 2} \right) - \left(\frac{1}{\ln 5} - \frac{1}{\ln 2} \right) \\
 &= \frac{16}{5 \ln 5} - \frac{1}{2 \ln 2}
 \end{aligned}$$



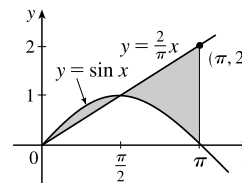
$$\begin{aligned}
 35. A &= \int_0^1 (e^{3x} - e^x) dx = \left[\frac{1}{3} e^{3x} - e^x \right]_0^1 \\
 &= \left(\frac{1}{3} e^3 - e \right) - \left(\frac{1}{3} - 1 \right) = \frac{1}{3} e^3 - e + \frac{2}{3}
 \end{aligned}$$



$$\begin{aligned}
 36. A &= \int_{-2}^0 (e^{-x} - e^x) dx + \int_0^1 (e^x - e^{-x}) dx \\
 &= [-e^{-x} - e^x]_{-2}^0 + [e^x + e^{-x}]_0^1 \\
 &= (-1 - 1) - (-e^2 - e^{-2}) \\
 &\quad + (e + e^{-1}) - (1 + 1) \\
 &= e^2 + e + e^{-1} + e^{-2} - 4
 \end{aligned}$$



$$\begin{aligned}
 37. \int_0^\pi \left| \sin x - \frac{2}{\pi} x \right| dx &= \int_0^{\pi/2} \left(\sin x - \frac{2}{\pi} x \right) dx + \int_{\pi/2}^\pi \left(\frac{2}{\pi} x - \sin x \right) dx \\
 &= \left[-\cos x - \frac{x^2}{\pi} \right]_0^{\pi/2} + \left[\frac{x^2}{\pi} + \cos x \right]_{\pi/2}^\pi \\
 &= -\frac{\pi}{4} + 1 + (\pi - 1) - \frac{\pi}{4} = \frac{\pi}{2}
 \end{aligned}$$



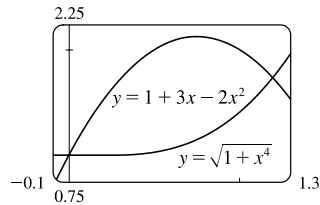
$$38. \text{ Let } f(x) = \sqrt{1+x^3} - (1-x), \Delta x = \frac{2-0}{4} = \frac{1}{2}.$$

$$\begin{aligned}
 A &= \int_0^2 \left[\sqrt{1+x^3} - (1-x) \right] dx \\
 &\approx \frac{1}{2} \left[f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) + f\left(\frac{5}{4}\right) + f\left(\frac{7}{4}\right) \right] \\
 &= \frac{1}{2} \left[\left(\frac{\sqrt{65}}{8} - \frac{3}{4} \right) + \left(\frac{\sqrt{91}}{8} - \frac{1}{4} \right) \right. \\
 &\quad \left. + \left(\frac{3\sqrt{21}}{8} + \frac{1}{4} \right) + \left(\frac{\sqrt{407}}{8} + \frac{3}{4} \right) \right] \\
 &= \frac{1}{16} \left(\sqrt{65} + \sqrt{91} + 3\sqrt{21} + \sqrt{407} \right) \approx 3.22
 \end{aligned}$$

$$39. \text{ Let } f(x) = x - x \tan x, \text{ and } \Delta x = \frac{\pi/4 - 0}{4} = \frac{\pi}{16}. \text{ Then}$$

$$\begin{aligned}
 A &= \int_0^{\pi/4} (x - x \tan x) dx \\
 &\approx \frac{\pi}{16} \left[f\left(\frac{\pi}{32}\right) + f\left(\frac{3\pi}{32}\right) + f\left(\frac{5\pi}{32}\right) + f\left(\frac{7\pi}{32}\right) \right] \\
 &\approx \frac{\pi}{16} \left[\frac{\pi}{32} \left(1 - \tan \frac{\pi}{32} \right) + \frac{3\pi}{32} \left(1 - \tan \frac{3\pi}{32} \right) \right. \\
 &\quad \left. + \frac{5\pi}{32} \left(1 - \tan \frac{5\pi}{32} \right) + \frac{7\pi}{32} \left(1 - \tan \frac{7\pi}{32} \right) \right] \\
 &= \frac{\pi^2}{512} \left[16 - \tan \frac{\pi}{32} - 3 \tan \frac{3\pi}{32} - 5 \tan \frac{5\pi}{32} - 7 \tan \frac{7\pi}{32} \right] \\
 &\approx 0.1267
 \end{aligned}$$

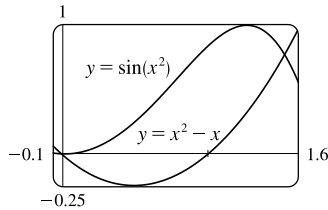
40.



From the graph, we see that the curves intersect at $x = 0$ and at $x \approx 1.19$, with $1 + 3x - 2x^2 > \sqrt{1 + x^4}$ on $(0, 1.19)$. So, using the Midpoint Rule with $f(x) = 1 + 3x - 2x^2 - \sqrt{1 + x^4}$ on $[0, 1.19]$ with $n = 4$, we calculate the approximate area between the curves:

$$\begin{aligned} A &\approx \int_0^{1.19} (1 + 3x - 2x^2 - \sqrt{1 + x^4}) dx \\ &\approx \frac{1.19}{4} [f(\frac{1.19}{8}) + f(\frac{3 \cdot 1.19}{8}) \\ &\quad + f(\frac{5 \cdot 1.19}{8}) + f(\frac{7 \cdot 1.19}{8})] \approx 0.83 \end{aligned}$$

41.

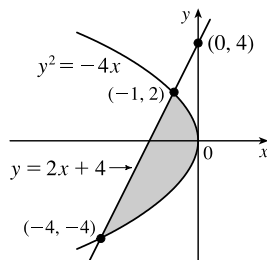


From the graph, we see that the curves intersect at $x = 0$ and at $x \approx 1.51$, with $\sin(x^2) > x^2 - x$ on $(0, 1.51)$. So, using the Midpoint Rule with $f(x) = \sin(x^2) - x^2 + x$ on $(0, 1.51)$ with $n = 4$, we calculate that the area between the curves is

$$\begin{aligned} A &\approx \int_0^{1.51} [\sin(x^2) - (x^2 - x)] dx \\ &\approx \frac{1.51}{4} [f(\frac{1.51}{8}) + f(\frac{3 \cdot 1.51}{8}) \\ &\quad + f(\frac{5 \cdot 1.51}{8}) + f(\frac{7 \cdot 1.51}{8})] \approx 0.81 \end{aligned}$$

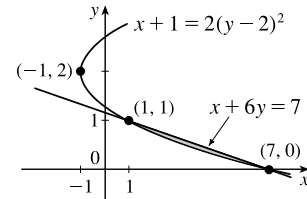
$$\begin{aligned} 42. (a) A &= \int_{-4}^{-1} [(2x + 4) + \sqrt{-4x}] dx + \int_{-1}^0 2\sqrt{-4x} dx \\ &= [x^2 + 4x]_{-4}^{-1} + 2 \int_{-4}^{-1} \sqrt{-x} dx + 4 \int_{-1}^0 \sqrt{-x} dx \\ &= (-3 - 0) + 2 \int_1^4 \sqrt{u} du + 4 \int_0^1 \sqrt{u} du \quad (u = -x) \\ &= -3 + [\frac{4}{3} u^{3/2}]_1^4 + [\frac{8}{3} u^{3/2}]_0^1 \\ &= -3 + \frac{28}{3} + \frac{8}{3} = 9 \end{aligned}$$

$$\begin{aligned} (b) A &= \int_{-4}^2 [-\frac{1}{4}y^2 - (\frac{1}{2}y - 2)] dy \\ &= [-\frac{1}{12}y^3 - \frac{1}{4}y^2 + 2y]_{-4}^2 \\ &= (-\frac{2}{3} - 1 + 4) - (-\frac{16}{3} - 4 - 8) = 9 \end{aligned}$$

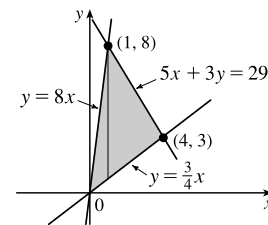


$$\begin{aligned} 43. (a) A &= \int_1^7 \left[\left(\frac{7}{6} - \frac{1}{6}x \right) - \left(2 - \sqrt{\frac{1}{2}(x+1)} \right) \right] dx \\ &= \int_1^7 \left[-\frac{5}{6} - \frac{1}{6}x + \frac{1}{\sqrt{2}}(x+1)^{1/2} \right] dx \\ &= \left[-\frac{5}{6}x - \frac{1}{12}x^2 + \frac{1}{\sqrt{2}} \cdot \frac{2}{3}(x+1)^{3/2} \right]_1^7 \\ &= -5 - 4 + \frac{\sqrt{2}}{3} (8\sqrt{8} - 2\sqrt{2}) = \frac{1}{3} \end{aligned}$$

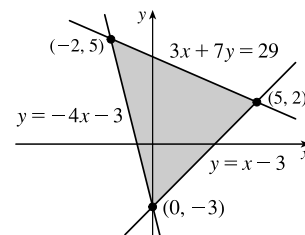
$$\begin{aligned} (b) A &= \int_0^1 ((7 - 6y) - [2(y - 2)^2 - 1]) dy \\ &= \int_0^1 (-2y^2 + 2y) dy = \left[-\frac{2}{3}y^3 + y^2 \right]_0^1 \\ &= -\frac{2}{3} + 1 = \frac{1}{3} \end{aligned}$$



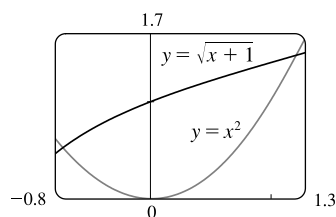
$$\begin{aligned} 44. A &= \int_0^1 (8x - \frac{3}{4}x) dx + \int_1^4 \left[(-\frac{5}{3}x + \frac{29}{3}) - \frac{3}{4}x \right] dx \\ &= \frac{29}{4} \int_0^1 x dx + \int_1^4 (-\frac{29}{12}x + \frac{29}{3}) dx \\ &= \frac{29}{4} [\frac{1}{2}x^2]_0^1 - \frac{29}{12} [\frac{1}{2}x^2 - 4x]_1^4 \\ &= \frac{29}{8} - \frac{29}{12} (-8 - \frac{1}{2} + 4) = \frac{29}{2} \end{aligned}$$



$$\begin{aligned} 45. A &= \int_{-2}^0 \left[(-\frac{3}{7}x + \frac{29}{7}) - (-4x - 3) \right] dx \\ &\quad + \int_0^5 \left[(-\frac{3}{7}x + \frac{29}{7}) - (x - 3) \right] dx \\ &= \int_{-2}^0 [\frac{25}{7}x + \frac{50}{7}] dx + \int_0^5 [-\frac{10}{7}x + \frac{50}{7}] dx \\ &= [\frac{25}{7}(\frac{1}{2}x^2) + \frac{50}{7}x]_{-2}^0 + [-\frac{5}{7}x^2 + \frac{50}{7}x]_0^5 \\ &= \frac{25}{7}(0 - 2) + \frac{50}{7}(0 + 2) \\ &\quad - \frac{5}{7}(25 - 0) + \frac{50}{7}(5 - 0) = 25 \end{aligned}$$



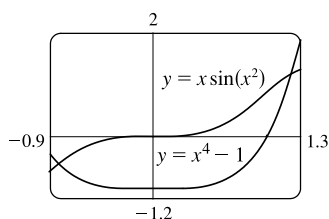
46.



From the graph, we see that the curves intersect at $x \approx -0.72$ and at $x \approx 1.22$, with $\sqrt{x+1} > x^2$ on $[-0.72, 1.22]$. So the area between the curves is

$$\begin{aligned} A &\approx \int_{-0.72}^{1.22} (\sqrt{x+1} - x^2) dx \\ &= \left[\frac{2}{3} (x+1)^{3/2} - \frac{1}{3} x^3 \right]_{-0.72}^{1.22} \\ &\approx 1.38 \end{aligned}$$

47.



From the graph, we see that the curves intersect at $x \approx -0.83$ and $x \approx 1.22$, with $x \sin(x^2) > x^4 - 1$ on $[-0.83, 1.22]$. So the area between the curves is

$$\begin{aligned} A &\approx \int_{-0.83}^{1.22} [x \sin(x^2) - (x^4 - 1)] dx \\ &= \left[-\frac{1}{2} \cos(x^2) - \frac{1}{5} x^5 + x \right]_{-0.83}^{1.22} \\ &\approx 1.78 \end{aligned}$$

48. A typical graphing calculator solution is as follows. Assign X^2 to $Y1$ ($y_1 = x^2$) and $\text{Exp}(-X^2)$ to $Y2$ ($y_2 = e^{-x^2}$). Graph the functions and find (and store) the x -coordinates of the points of intersection. In this case, we have some symmetry, so we need to find only one point of intersection. Store $x \approx 0.75308916$ in memory location B. Now use the appropriate integration command to approximate the area:

$$\begin{aligned} A &= 2 \int_0^B (y_2 - y_1) dx \\ &= 2 * \text{Int}(Y2 - Y1, X, 0, B) \\ &\approx 0.979263 \end{aligned}$$

